UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 24/09/2019*

PROBLEM

We observe N samples of a realization of the random process $X[n] = A + B \cdot s[n] + W[n]$, for n=0, 1,…, N-1. where A and B are two unknown deterministic parameters that we want to estimate, and $\{W[n]; n = 0,1,\cdots,N-1\}$ are independent and identically distributed (IID) random variables modeling additive noise. Each noise sample is Gaussian distributed, zero mean and with known variance σ_W^2. The signal $s[n]$ is a bipolar pulse defined as follows:

$$s[n] = \begin{cases} 2, & 0 \le n < N/2 \\ -2 & N/2 \le n \le N-1 \\ 0, & elsewhere \end{cases}$$

(1) Derive the log-likelihood function (LLF) for the estimate of the parameters A and B.

(2) Derive the joint maximum likelihood (ML) estimator of the parameters A and B.

(3) Derive the bias and the mean square error (MSE) of the ML estimators of A and B, and assess whether they are unbiased and consistent or not.

(4) Derive the joint CRB of the two estimators and assess whether they are efficient or not.

(5) Plot the MSE as a function of σ_W^2 and find the minimum value of N which guarantees an MSE lower than 10^{-4} for both of the two ML estimators when the noise power is σ_W^2 =2.

Solution

(1) Derive the log-likelihood function (LLF) for the estimate of the parameters *A* and *B*.

Let's define $X_n \triangleq X[n]$ and $s_n \triangleq s[n]$. The pdf of the observed data vector is given by:

$X_n \in N\left(A + Bs_n, \sigma_W^2\right), \quad \left\{X_n\right\}_{n=0}^{N-1}$ mutually independent $\Rightarrow \quad \mathbf{X} \in N\left(A\mathbf{1} + B\mathbf{s}, \sigma_W^2 \mathbf{I}\right)$

The pdf in scalar format is:

$$f_{\mathbf{X}}(\mathbf{x}; A, B) = \prod_{n=0}^{N-1} f_X(x_n; A, B) = \left(\frac{1}{\sqrt{2\pi\sigma_W^2}}\right)^N \exp\left(-\frac{1}{2\sigma_W^2}\sum_{n=0}^{N-1}\left(x_n - A - B\cdot s_n\right)^2\right)$$

$$LLF(A,B) = \ln f_{\mathbf{X}}(\mathbf{x}; A, B) = k - \frac{1}{2\sigma_W^2}\sum_{n=0}^{N-1}\left(x_n - A - B\cdot s_n\right)^2$$

In vector format:

$$f_{\mathbf{X}}(\mathbf{x}; A, B) = \frac{1}{\left(2\pi\sigma_W^2\right)^{N/2}} \exp\left(-\frac{1}{2\sigma_W^2}(\mathbf{x} - A\mathbf{1} - B\mathbf{s})^T(\mathbf{x} - A\mathbf{1} - B\mathbf{s})\right)$$

$$LLF(A,B) = \ln f_{\mathbf{X}}(\mathbf{x}; A, B) = k - \frac{1}{2\sigma_W^2}(\mathbf{x} - A\mathbf{1} - B\mathbf{s})^T(\mathbf{x} - A\mathbf{1} - B\mathbf{s})$$

(2) Derive the joint maximum likelihood (ML) estimator of the parameters *A* and *B*.

To derive the two ML estimators we can use the LLF either in scalar or in vector notations. Let's use the vector's one:

$$LLF(A,B) = \ln f_{\mathbf{X}}(\mathbf{x}; A, B) = k - \frac{1}{2\sigma_W^2}(\mathbf{x} - A\mathbf{1} - B\mathbf{s})^T(\mathbf{x} - A\mathbf{1} - B\mathbf{s})$$

$$= k - \frac{1}{2\sigma_W^2}(\mathbf{x}^T\mathbf{x} - A\mathbf{1}^T\mathbf{x} - B\mathbf{s}^T\mathbf{x} - A\mathbf{x}^T\mathbf{1} + A^2\mathbf{1}^T\mathbf{1} + AB\mathbf{s}^T\mathbf{1} - B\mathbf{x}^T\mathbf{s} + AB\mathbf{1}^T\mathbf{s} + B^2\mathbf{s}^T\mathbf{s})$$

Taking into account that $\mathbf{1}^T\mathbf{1} = N, \mathbf{s}^T\mathbf{s} = 4N, \mathbf{s}^T\mathbf{1} = \mathbf{1}^T\mathbf{s} = 0$, we get:

$$LLF(A,B) = k - \frac{1}{2\sigma_W^2}(\mathbf{x}^T\mathbf{x} - A\mathbf{1}^T\mathbf{x} - B\mathbf{s}^T\mathbf{x} - A\mathbf{x}^T\mathbf{1} + NA^2 - B\mathbf{x}^T\mathbf{s} + 4NB^2)$$

The two joint ML estimators are derived by solving the likelihood equations:

$$\begin{cases} \dfrac{dLLF(A,B)}{dA} = -\dfrac{1}{2\sigma_W^2}(-\mathbf{x}^T\mathbf{1}-\mathbf{1}^T\mathbf{x}+2NA) = -\dfrac{1}{\sigma_W^2}(NA-\mathbf{x}^T\mathbf{1}) = 0 \\ \\ \dfrac{dLLF(A,B)}{dB} = -\dfrac{1}{2\sigma_W^2}(-\mathbf{x}^T\mathbf{s}-\mathbf{s}^T\mathbf{x}+8NB) = -\dfrac{1}{\sigma_W^2}(4NB-\mathbf{x}^T\mathbf{s}) = 0 \end{cases}$$

$$\begin{cases} \hat{A}_{ML} = \dfrac{1}{N}\mathbf{x}^T\mathbf{1} = \dfrac{1}{N}\sum_{n=0}^{N-1} X[n] \\ \\ \hat{B}_{ML} = \dfrac{1}{4N}\mathbf{x}^T\mathbf{s} = \dfrac{1}{4N}\sum_{n=0}^{N-1} s[n]X[n] = \dfrac{1}{2N}\left(\sum_{n=0}^{N/2-1} X[n] - \sum_{n=N/2}^{N-1} X[n]\right) \end{cases}$$

(3) Derive the bias and the mean square error (MSE) of the ML estimators of *A* and *B*, and assess whether they are unbiased and consistent or not.

Since both the estimates are linear functions of the Gaussian distributed data vector **X**, they are Gaussian distributed, and consequently the estimation error is Gaussian distributed, with mean and variance given by:

$$E\left\{\hat{A}_{ML}\right\} = E\left\{\frac{1}{N}\mathbf{x}^T\mathbf{1}\right\} = \frac{1}{N}\left(E\{\mathbf{x}\}\right)^T\mathbf{1} = \frac{1}{N}\left(A\mathbf{1}+B\mathbf{s}\right)^T\mathbf{1} = \frac{A}{N}\mathbf{1}^T\mathbf{1} = A$$

$$E\left\{\hat{B}_{ML}\right\} = E\left\{\frac{1}{4N}\mathbf{x}^T\mathbf{s}\right\} = \frac{1}{4N}\left(E\{\mathbf{x}\}\right)^T\mathbf{s} = \frac{1}{4N}\left(A\mathbf{1}+B\mathbf{s}\right)^T\mathbf{1} = \frac{B}{4N}\mathbf{s}^T\mathbf{s} = B$$

Hence, both the estimators are unbiased.

$$MSE\left\{\hat{A}_{ML}\right\} = E\left\{\left(A-\hat{A}_{ML}\right)^2\right\} = \mathrm{var}\left\{\hat{A}_{ML}\right\} = \mathrm{var}\left\{\frac{1}{N}\sum_{n=0}^{N-1} X[n]\right\} = \frac{1}{N^2}\sum_{n=0}^{N-1}\mathrm{var}\left\{X[n]\right\} = \frac{N\sigma_W^2}{N^2} = \frac{\sigma_W^2}{N}$$

$$MSE\left\{\hat{B}_{ML}\right\} = E\left\{\left(B-\hat{B}_{ML}\right)^2\right\} = \text{var}\left\{\hat{B}_{ML}\right\} = \text{var}\left\{\frac{1}{4N}\sum_{n=0}^{N-1} s[n]X[n]\right\} = \frac{1}{16N^2}\sum_{n=0}^{N-1}\text{var}\left\{s[n]X[n]\right\}$$
$$= \frac{1}{16N^2}\sum_{n=0}^{N-1} s^2[n]\,\text{var}\left\{X[n]\right\}\frac{1}{16N^2}\sum_{n=0}^{N-1} s^2[n]\sigma_W^2 = \frac{\sigma_W^2}{16N^2}\sum_{n=0}^{N-1} s^2[n] = \frac{\sigma_W^2}{16N^2}4N = \frac{\sigma_W^2}{4N}$$

$\lim_{N\to\infty} MSE\left\{\hat{A}_{ML}\right\} = 0, \quad \lim_{N\to\infty} MSE\left\{\hat{B}_{ML}\right\} = 0$. Hence, both the estimators are consistent.

(4) Derive the joint CRB of the two estimators and assess whether they are efficient or not.

$$\begin{cases} \dfrac{dLLF(A,B)}{dA} = -\dfrac{1}{\sigma_W^2}(NA - \mathbf{x}^T\mathbf{1}) \\ \\ \dfrac{dLLF(A,B)}{dB} = -\dfrac{1}{\sigma_W^2}(4NB - \mathbf{x}^T\mathbf{s}) \end{cases} \Rightarrow \begin{cases} \dfrac{d^2LLF(A,B)}{dA^2} = -\dfrac{N}{\sigma_W^2} \\ \\ \dfrac{d^2LLF(A,B)}{dB^2} = -\dfrac{4N}{\sigma_W^2} \end{cases}$$

Moreover, $\dfrac{d^2LLF(A,B)}{dAdB} = 0$, so the FIM is diagonal and the two CRBs are given by:

$$CRB(A) = \frac{1}{-E\left\{\dfrac{d^2LLF(A,B)}{dA^2}\right\}} = \frac{\sigma_W^2}{N} \quad \text{and} \quad CRB(B) = \frac{1}{-E\left\{\dfrac{d^2LLF(A,B)}{dB^2}\right\}} = \frac{\sigma_W^2}{4N}$$

Hence, both the estimators are efficient.

(5) Plot the MSE as a function of σ_W^2 and find the minimum value of N which guarantees an MSE lower than 10^{-3} for both of the two ML estimators when the noise power is $\sigma_W^2 = 2$.

Since the estimator of A is the one with the greatest MSE between the two, we have to impose the constraint on it:

$$MSE\left\{\hat{A}_{ML}\right\} = \frac{\sigma_W^2}{N} < 10^{-4} \quad \Rightarrow \quad \frac{2}{N} < 10^{-4} \quad \Rightarrow \quad N > 2\cdot 10^4$$